

08.1

A study of cells from a human embryo found that $2.66 \times 10^{-7} \%$ of the genome of one cell had changed due to mutation.

You can assume that:

- the genome contains 3×10^9 nucleotides
- the mean rate of mutation is 1.14 mutations per cell cycle
- the cell cycle lasts 24 hours.

Use this information to calculate the age of the embryo in days.

Show your working.

[2 marks]

Answer _____ days

The genome of the bacterium *Escherichia coli* is approximately 4.6 million base pairs in length and contains around 4000 genes.

08.2

The analysis of the genome of *E. coli* allows the proteome to be determined more easily than that of eukaryotes.

Explain why.

[2 marks]

Question 8 continues on the next page

0 8 . 3

Determining the proteomes of bacteria has helped provide protection for individuals and populations against some infectious diseases.

Explain how.

[2 marks]

Scientists investigated the relationship between one type of variable number tandem repeat (VNTR) and the risk of breast cancer occurring in women.

The scientists determined the risk of breast cancer occurring in women:

- with 2, 3 or 4 repeats of the VNTR
- **without** the VNTR.

They calculated the odds ratio (OR) of breast cancer occurring with 2, 3 or 4 repeats of the VNTR.

An OR is a measure of how strongly an outcome is associated with a risk factor.

The OR formula is:

risk of breast cancer occurring **with** the VNTR : risk of breast cancer occurring **without** the VNTR

Table 5 shows the scientists' results.

Table 5

Number of VNTR repeats	Odds ratio
2	0.55 : 1
3	1.58 : 1
4	1.94 : 1

A student concluded that the number of repeats of the VNTR is associated with the risk of breast cancer occurring in women.

[4 marks]

[illegible]

Turn over for the next question

